

5a	As <u>temperature (of the filament) increases</u> with V the <u>resistance of the lamp increases</u> hence V/I ratio increases.	A	M1 A1
bi	From I - V graph, find the common current when the individual p.d. of each lamp sum up to 12 V. $I = 0.40 \text{ A } (\pm 0.01 \text{ A})$	A	B1 B1

bii	Current is the same, but <u>p.d. across lamp B is higher than that of lamp A</u> OR resistance of lamp B is higher than resistance of lamp A Using $P = IV$, lamp <u>B</u> will have <u>greater</u> power dissipation.	A	M1 A1
c	From bi, $V_A = 4.0 \text{ V}$, $V_B = 8.0 \text{ V}$ For zero current in the galvanometer G, $k(XJ) = 4V$ where $k = 12/0.80$ $XJ = (4/12) \times 0.80$ $= 0.27 \text{ (0.267) m}$ OR For zero current in the galvanometer G $XJ = [4/(4+8)] \times 0.80$ $= 0.27 \text{ (0.267) m}$	A	C1 A1 (C1) (A1)

6a	Each coil acts as a current carrying conductor in the magnetic field of the	A	
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